

5) Obtain partial diff. eqn by eliminating arbitrary functions from the following

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(a) $z = f_1(y+2x) + f_2(y-3x)$

Sol:- Partially Given eqn $z = f_1(y+2x) + f_2(y-3x) \rightarrow (1)$
Diff/- eqn (1) w.r.t 'x' we get

$$\frac{\partial z}{\partial x} = f_1'(y+2x)(2) + f_2'(y-3x)(-3)$$

$$\frac{\partial z}{\partial x} = 2f_1'(y+2x) - 3f_2'(y-3x) \rightarrow (2)$$

Partially Diff/- with respect to 'y', we get

$$\frac{\partial z}{\partial y} = f_1'(y+2x)(1) + f_2'(y-3x)(1) \rightarrow (3)$$

from (1), (2) & (3) we can't eliminate functions. So, we require

second order p.d.e. w.r.t 'x', we get

$$\frac{\partial}{\partial x} (\text{Eqn (2)}) = \frac{\partial}{\partial x} \left[\frac{\partial z}{\partial x} \right] = 2f_1''(y+2x)(2) - 3f_2''(y-3x)(-3)$$

$$\frac{\partial^2 z}{\partial x^2} = 4f_1''(y+2x) + 9f_2''(y-3x) \rightarrow (4)$$

$$\frac{\partial}{\partial x} (\text{Eqn (3)}) = \frac{\partial}{\partial x} \left[\frac{\partial z}{\partial y} \right] = f_1''(y+2x)(2) + f_2''(y-3x)(-3)$$

$$\frac{\partial^2 z}{\partial x \partial y} = 2f_1''(y+2x) - 3f_2''(y-3x) \rightarrow (5)$$

Partially Diff/- Eqn (3) w.r.t 'y',

$$\frac{\partial}{\partial y} (\text{Eqn (3)}) \Rightarrow \frac{\partial}{\partial y} \left[\frac{\partial z}{\partial y} \right] = f_1''(y+2x) + f_2''(y-3x) \rightarrow (6)$$

Solving Eqn (4) & (5) we get

$$\text{Eqn (4)} \Rightarrow \frac{\partial^2 z}{\partial x^2} = 4f_1''(y+2x) + 9f_2''(y-3x)$$

$$\text{Eqn (5)} \times 2 \Rightarrow 2 \frac{\partial^2 z}{\partial x \partial y} = 4f_1''(y+2x) - 6f_2''(y-3x)$$

$$\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} = 15f_2''(y-3x)$$

$$\frac{1}{15} \left[\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} \right] = f_2''(y-3x) \rightarrow (7)$$

Subl- Eqn (7) in Eqn (4). we get

$$\frac{\partial^2 z}{\partial x^2} = 4 f_1''(y+2x) + \frac{9}{15} \left[\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} \right]$$

$$\frac{\partial^2 z}{\partial x^2} - \frac{3}{5} \left[\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} \right] = 4 f_1''(y+2x)$$

$$\frac{\partial^2 z}{\partial x^2} - \frac{3}{5} \frac{\partial^2 z}{\partial x^2} + \frac{6}{5} \left(\frac{\partial^2 z}{\partial x \partial y} \right) = 4 f_1''(y+2x)$$

$$\frac{2}{5} \frac{\partial^2 z}{\partial x^2} + \frac{6}{5} \left(\frac{\partial^2 z}{\partial x \partial y} \right) = 4 f_1''(y+2x)$$

$$\frac{2}{5} \left[\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} \right] = 4 f_1''(y+2x)$$

$$f_1''(y+2x) = \frac{1}{10} \left[\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} \right] \rightarrow (8)$$

Subl- (7) & (8) in Eqn (6)

$$\frac{\partial^2 z}{\partial y^2} = \frac{1}{10} \left[\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} \right] + \frac{1}{15} \left[\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} \right]$$

$$= \frac{3}{30} \left[\frac{\partial^2 z}{\partial x^2} + 3 \frac{\partial^2 z}{\partial x \partial y} \right] + \frac{2}{30} \left[\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial^2 z}{\partial x \partial y} \right]$$

$$= \frac{1}{30} \left[3 \frac{\partial^2 z}{\partial x^2} + 9 \frac{\partial^2 z}{\partial x \partial y} + 2 \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x \partial y} \right]$$

$$= \frac{1}{30} \left[5 \frac{\partial^2 z}{\partial x^2} + 5 \frac{\partial^2 z}{\partial x \partial y} \right] = \frac{1}{6} \left[\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} \right]$$

$$t = \frac{1}{6} [x+s] \Rightarrow 6t = x+s$$

$$x+s-6t=0$$

2t
b) $z = f(\sin x + \cos y)$

sol:- Given Eqn $z = f(\sin x + \cos y) \rightarrow (1)$

Diff/- Eqn (1) partially w.r.t x and y , we get

$$\frac{\partial z}{\partial x} = f'(\sin x + \cos y) \cdot \cos x$$

$$p = f'(\sin x + \cos y) \cos x \rightarrow (2)$$

$$\left\{ \begin{array}{l} \because \frac{1}{10} + \frac{1}{15} \\ 5 \overline{) 10, 15} \\ \underline{2, 3} \\ 5 \times 2 + 3 = 30 \end{array} \right.$$

$$11^y \quad \frac{\partial z}{\partial y} = f'(\sin x + \cos y) \cdot (-\sin y)$$

$$q = f'(\sin x + \cos y) (-\sin y) \rightarrow (3)$$

Dividing $\frac{\text{Eqn (2)}}{\text{Eqn (3)}} \Rightarrow \frac{p}{q} = \frac{f'(\sin x + \cos y) \cos x}{f'(\sin x + \cos y) (-\sin y)}$

$$\frac{p}{q} = \frac{\cos x}{-\sin y}$$

$$-p \sin y = q \cos x$$

$$\Rightarrow p \sin y + q \cos x = 0$$

which is the required partial diff/- eqn.

Home work Problems

* Form a partial diff/- eqn by eliminating the arbitrary functions from

(6) $z = f(x) + e^y g(x)$

sol:- Given eqn $z = f(x) + e^y g(x) \rightarrow (1)$

Diff/- eqn (1) partially w.r.t x & y .

$$\frac{\partial z}{\partial x} = f'(x) + e^y g'(x) \rightarrow (2) \quad \left| \quad \frac{\partial z}{\partial y} = 0 + e^y g(x) \right.$$

$$\frac{\partial z}{\partial y} = e^y g(x) \rightarrow (3)$$

Diff/- (3) partially w.r.t 'y', we have

$$\frac{\partial^2 z}{\partial y^2} = e^y g(x) \quad [\because \text{from Eqn (3)}]$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial z}{\partial y} \Rightarrow t = z \Rightarrow \boxed{t - z = 0}$$

Important note:- $z = f(x) + e^y g(x)$ is the general solution of $\frac{\partial^2 z}{\partial y^2} - \frac{\partial z}{\partial y} = 0$.
Contains two arbitrary functions. In general, the general solution of diff/- eqn of n^{th} order contains n arbitrary functions.

2times
7)

$$z = f(y/x)$$

sol:- we have $z = f[y/x] \rightarrow (1)$

Diff/- Eqn (1) partially w.r.t x and y , we get.

$$\frac{\partial z}{\partial x} = f' \left[\frac{y}{x} \right] \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right) \Rightarrow$$

$$p = f' \left(\frac{y}{x} \right) \left(-\frac{y}{x^2} \right) \rightarrow (2)$$

||^y $\frac{\partial z}{\partial y} = f' \left(\frac{y}{x} \right) \cdot \frac{\partial}{\partial y} \left(\frac{y}{x} \right)$

$$q = f' \left(\frac{y}{x} \right) \cdot \frac{1}{x} \rightarrow (3)$$

let us eliminate $f'(y/x)$ from Eqn (2) & Eqn (3)

$$\frac{\text{Eqn (2)}}{\text{Eqn (3)}} \Rightarrow \frac{p}{q} = \frac{f' \left(\frac{y}{x} \right) \left(-\frac{y}{x^2} \right)}{f' \left(\frac{y}{x} \right) \left(\frac{1}{x} \right)}$$

$$\frac{p}{q} = \frac{(-y/x^2)}{(1/x)} \Rightarrow \frac{p}{q} = -\frac{y}{x} \Rightarrow px = -qy$$

$$\therefore \boxed{px + qy = 0}$$

2times which is the required partial diff/- Eqn.

8) $z = xy + f(x^2 + y^2)$

sol:- Given Eqn $z = xy + f(x^2 + y^2) \rightarrow (1)$

Diff/- Eqn (1) partially w.r.t x & y

$$\frac{\partial z}{\partial x} = y(1) + f'(x^2 + y^2)(2x)$$

$$p = y + 2x f'(x^2 + y^2) \rightarrow (2)$$

||^y $\frac{\partial z}{\partial y} = x + f'(x^2 + y^2)(2y)$

$$q = x + 2y f'(x^2 + y^2) \rightarrow (3)$$

Dividing $\frac{\text{Eqn (2)}}{\text{Eqn (3)}} \Rightarrow \frac{p-y}{q-x} = \frac{2x f'(x^2 + y^2)}{2y f'(x^2 + y^2)}$

$$\frac{p-y}{z-x} = \frac{x}{y}$$

$$py - y^2 = qx - x^2$$

$$x^2 - y^2 = zx - py$$

which is the required solution.

2t

9) $z = f(x^2 + y^2 + z^2)$

Sol:- Given $z = f(x^2 + y^2 + z^2) \rightarrow (1)$

Diff- Eqn (1) partially w.r.t 'x' & 'y', we get

$$\frac{\partial z}{\partial x} = f'(x^2 + y^2 + z^2) (2x + 2z \frac{\partial z}{\partial x})$$

$p = f'(x^2 + y^2 + z^2) (2x + 2z p) \rightarrow (2) \quad \boxed{p = \frac{f'(x^2 + y^2 + z^2)}{2x + 2z p}}$

11y $\frac{\partial z}{\partial y} = f'(x^2 + y^2 + z^2) (2y + 2z \frac{\partial z}{\partial y})$

$$q = f'(x^2 + y^2 + z^2) (2y + 2z q) \rightarrow$$

$$\boxed{\frac{q}{2y + 2z q} = f'(x^2 + y^2 + z^2)} \rightarrow (3)$$

Solving Eqn (2) & (3) $\Rightarrow \frac{p}{2x + 2z p} = \frac{q}{2y + 2z q}$

$$2y p + 2z p q = 2x q + 2z p q$$

$$2y p + 2z p q - 2x q - 2z p q = 0$$

$$2y p - 2x q = 0$$

$$\boxed{py - qx = 0}$$

This is a partial diff- Eqn of first order

2t

10) $z = y^2 + 2f\left[\frac{1}{x} + \log y\right]$

Sol:- Given Eqn $z = y^2 + 2f\left[\frac{1}{x} + \log y\right] \rightarrow (1)$

Diff- Eqn (1) partially w.r.t 'x' & 'y', we get

$$\frac{\partial z}{\partial x} = 2f' \left[\frac{1}{x} + \log y \right] \left(-\frac{1}{x^2} \right)$$

$$p = 2f' \left[\frac{1}{x} + \log y \right] \left(-\frac{1}{x^2} \right)$$

$$-px^2 = 2f' \left[\frac{1}{x} + \log y \right] \rightarrow (2)$$

$$\frac{\partial z}{\partial y} = 2y + 2f' \left[\frac{1}{x} + \log y \right] \left(\frac{1}{y} \right)$$

$$q = 2y + 2f' \left[\frac{1}{x} + \log y \right] \left(\frac{1}{y} \right)$$

$$q - 2y = 2f' \left(\frac{1}{x} + \log y \right) \left(\frac{1}{y} \right)$$

$$(q - 2y)y = 2f' \left(\frac{1}{x} + \log y \right)$$

$$qy - 2y^2 = 2f' \left(\frac{1}{x} + \log y \right) \rightarrow (3)$$

Solving Eqn (2) & (3)

$$-px^2 = qy - 2y^2$$

$$px^2 + qy - 2y^2 = 0$$

$$\boxed{px^2 + qy = 2y^2}$$

Lagrange's Linear Equation

A linear partial diff eqn of first order, involving dependent variable z and two independent variables x & y , of the form $Pp + Qq = R$.

where P, Q, R are functions of x, y, z is called "Lagrange's Linear Equation".

→ The Lagrange's solution of the linear eqn is of the form

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$